SSTAC Survey Planner

User Manual & Technical Overview

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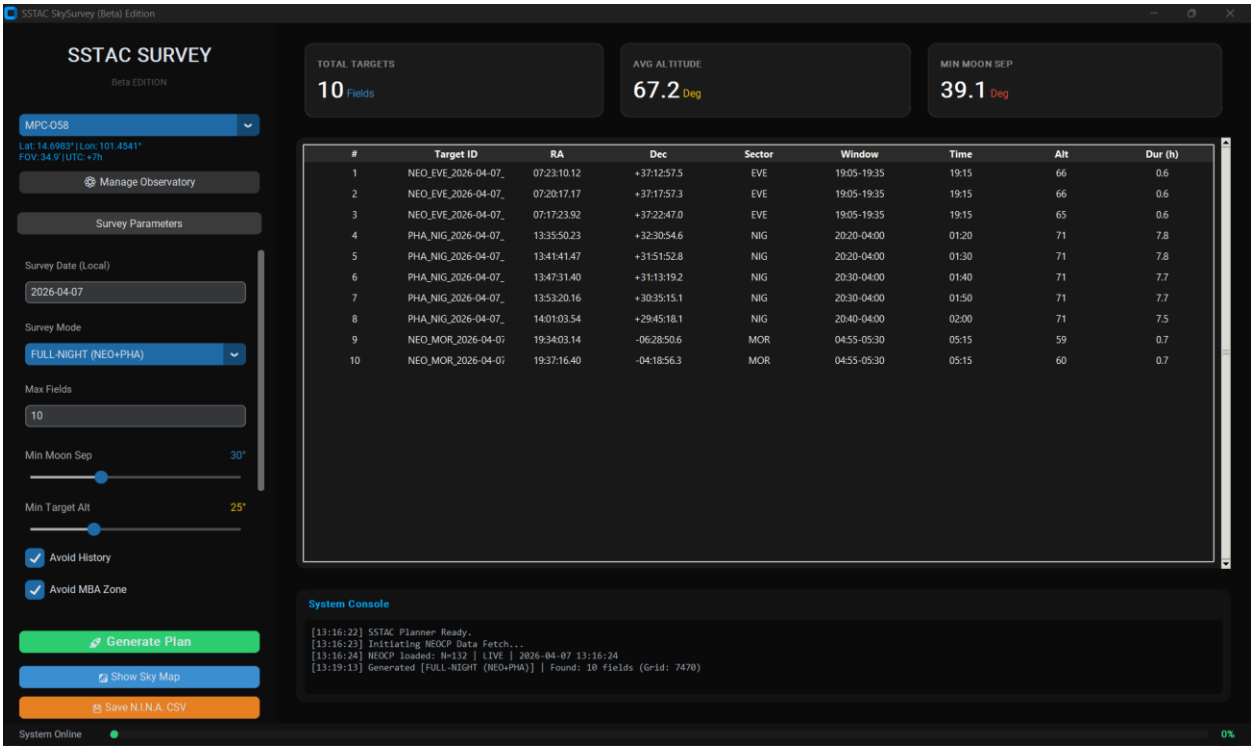

Figure 1: SSTAC Survey Planner

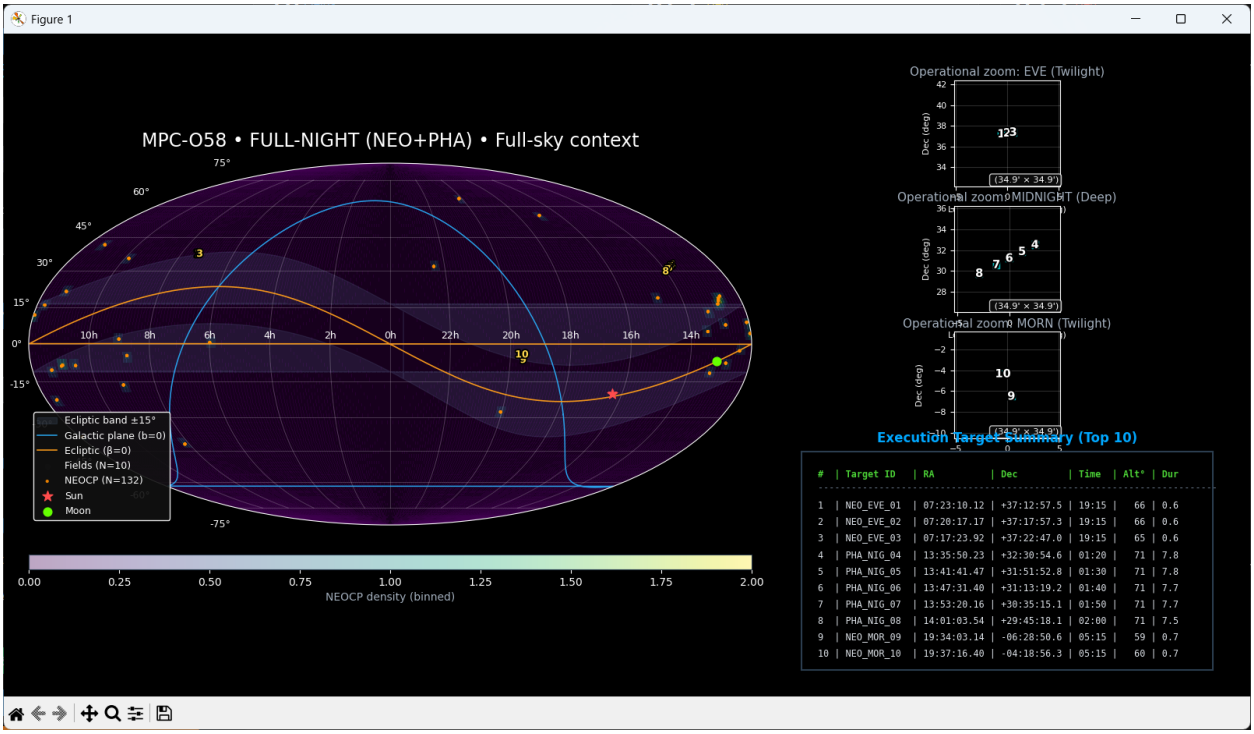

Figure 2: Sky Map for Full-Night (NEO - PHA) mode

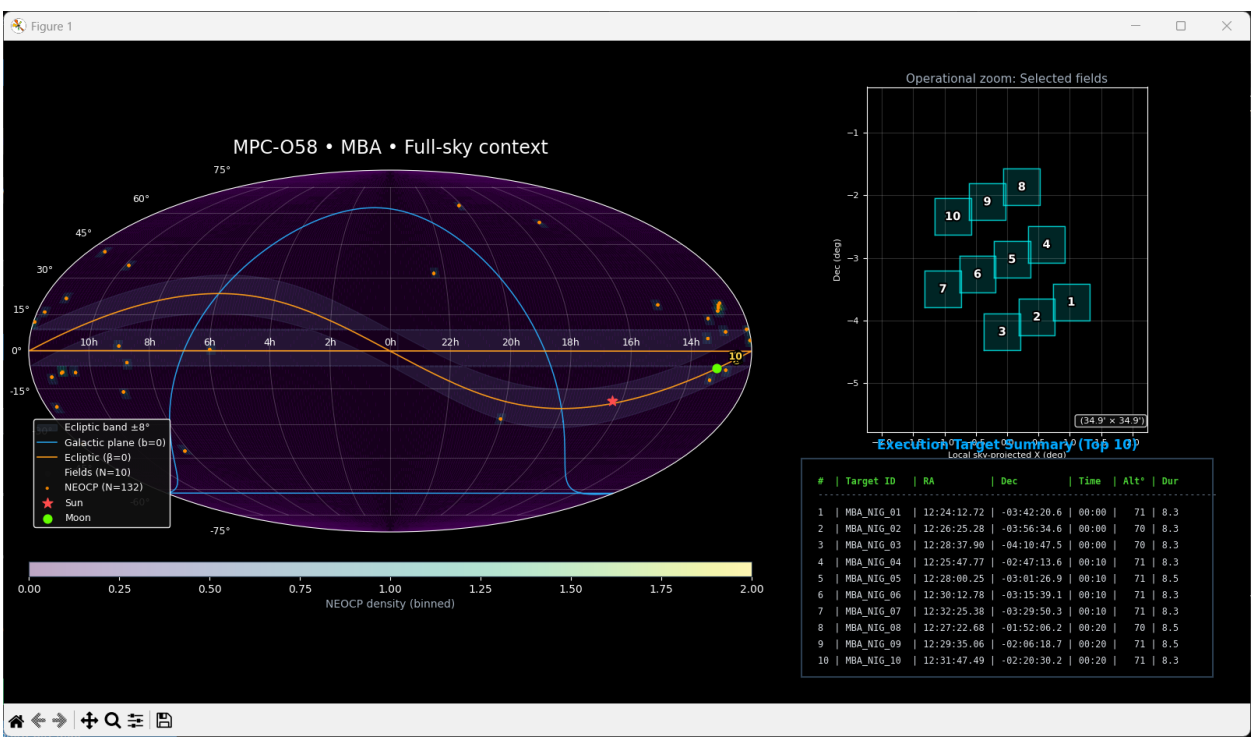

Figure 3: Sky Map for MBA mode

1. Introduction

SSTAC Survey Planner is a tool under **Small Survey Telescope for Asteroid and Comet (SSTAC) Project,** it is a geometry-driven sky survey planning tool designed for small-aperture observatories aiming to detect asteroids and comets. Unlike traditional sky survey approaches that rely on uniform sky coverage, SSTAC prioritizes observational efficiency by selecting fields based on:

- Solar elongation geometry
- Twilight observing windows
- Ecliptic latitude distribution
- Moon avoidance
- Galactic plane avoidance
- Field overlap control
- Optional NEOCP weighting

The system is optimized for **operational usability** rather than full physical simulation, making it particularly suitable for amateur and semi-professional observatories.

2. Conceptual Framework

2.1 Core Philosophy

SSTAC is based on the principle that:

“Not all parts of the sky are equally valuable for discovery at a given time.”

The planner identifies **“sweet spots”** in the sky where the probability of detecting moving objects is higher due to geometric and observational conditions.

2.2 Survey Strategy Model

The planner operates using a **three-layer model**:

Layer 1: Geometry-Based Candidate Generation

Fields are generated based on relative positions between:

- Sun
- Observer
- Ecliptic plane

Layer 2: Visibility Constraints

Fields are filtered using:

- Altitude
- Twilight window
- Moon separation
- Galactic latitude

Layer 3: Heuristic Scoring

Each field is ranked using a merit function:

$$Score = f(\text{twilight}) \times f(\text{elongation}) \times f(\text{altitude})$$

This score reflects **observational suitability**, not physical discovery probability.

3. Survey Modes

3.1 Mode Summary Table

Mode	Description	Geometry Focus	Use Case
MBA	Main-Belt Asteroids	Opposition (≈180° elongation)	Stable detection / baseline survey
NEO	Near-Earth Objects	Twilight (60°–90° elongation)	High-risk discovery
COMET-TWILIGHT	Sunward comet search	30°–90° elongation	Comet discovery
PHA-MIDNIGHT	Non-MBA objects	Off-ecliptic	Reduce MBA contamination
FULL-NIGHT (NEO+MBA)	Hybrid	Twilight + opposition	Balanced survey
FULL-NIGHT (COMET+MBA)	Hybrid	Sunward + opposition	Comet + MBA
FULL-NIGHT (NEO+PHA)	Hybrid	Twilight + off-ecliptic	NEO-focused
FULL-NIGHT (COMET+PHA)	Hybrid	Sunward + off-ecliptic	Clean comet search

3.2 Mode Descriptions

MBA (Main-Belt Asteroids)

- Focus: Near opposition
- Narrow ecliptic latitude band
- Highest object density
- Best for system validation and consistent detection

NEO (Near-Earth Objects)

- Focus: Twilight regions
- Solar elongation ~60°–90°
- Split into:
    - Evening (EVE)
    - Morning (MORN)
- High discovery potential, lower success rate

COMET-TWILIGHT

- Focus: Sunward region
- Wider latitude coverage
- Targets comet-like objects and unusual orbits

PHA-MIDNIGHT

- Focus: Away from ecliptic
- Reduces main-belt contamination
- Useful for cleaner candidate sets

FULL-NIGHT Modes

- Combine twilight + midnight strategies
- Conceptual structure:

Evening Twilight ⟶ Midnight Fields ⟶ Morning Twilight

Note:

Current implementation is **field-based hybrid**, not full scheduler optimization.

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4. Parameter Configuration

4.1 Core Parameters

Min Moon Separation (deg)

Minimum angular distance from Moon.

- Recommended: ≥ 30°
- Higher values reduce background noise

Min Target Altitude (deg)

Minimum elevation above horizon.

Value	Interpretation
20°	Minimum usable
25°	Recommended
30°	Conservative

Max Fields

Number of fields to generate.

- Typical:
    - 5–10 (small setup)
    - 15–25 (full-night survey)

4.2 Operational Filters

Avoid History

- Prevents re-observing previous fields
- Recommendation for multi-night surveys
- Disable if too few fields are generated

Avoid MBA Zone

- Removes ecliptic-dense region
- Enabled automatically for NEO/COMET
- Disabled for MBA modes

Avoid Galactic Plane

- Excludes dense star regions
- Recommend: ON

Overlap Control

- Prevents redundant fields
- Typical: 10–15%

NEOCP Weighting

- Prioritizes regions with known candidate density
- Should be used as soft weighting, not absolute filter

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5. Workflow

5.1 Pre-Observation

1. Select observatory profile
2. Set observing date

3. Choose survey mode
4. Configure parameters
5. Generate plan
6. Review sky map
7. Export to scheduler (e.g., NINA)

5.2 During Observation

- Monitor:
    - Field timing vs twilight window
    - Moon position
    - Sky conditions
- Adjust plan if necessary

5.3 Post-Observation

1. Run detection pipeline
2. Identify moving objects
3. Cross-match known objects
4. Record:
    - Candidate count
    - Faintest detected magnitude
5. (Optional) Integrate sky quality metrics

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6. Best Practices

6.1 Strategy Selection

Goal	Recommended Mode
System testing	MBA
Balanced survey	FULL-NIGHT (NEO+MBA)
Discovery attempt	NEO / COMET
Clean candidates	PHA

6.2 Operational Tips

- Start with MBA before advanced modes
- Use twilight modes only after system is stable
- Avoid over-restrictive filters
- Monitor field quality, not just quantity
- Always validate with known-object recovery

6.3 Parameter Tuning

- If too few fields:
    - Reduce Moon separation
    - Disable history
    - Lower altitude slightly
- If too many noisy detections:
    - Enable Galactic avoidance
    - Increase Moon separation

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7. Limitations

SSTAC is not a full physical simulation tool.

It does NOT include:

- SNR modeling
- Limiting magnitude physics
- Atmospheric extinction modeling
- Trailing loss modeling
- Orbital population simulation
- Full-night scheduling optimization

Important Note

The scoring system represents:

Observational suitability, not discovery probability.

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8. Intended Use

SSTAC is designed as:

- Field planning tool
- Survey support system
- Observational decision aid

It is NOT:

- Detection software
- Photometric pipeline
- Orbit determination system

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9. Integration with Survey Systems

Recommended integration:

SSTAC ⟶ Imaging ⟶ Detection (Tycho) ⟶ Validation ⟶ Logging

Optional extension:

- Sky quality estimation
- Candidate database
- Multi-night tracking

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10. Version Status

Current release should be considered:

Beta / Observatory Use Tool

Suitable for:

- Amateur observatories
- Small survey projects
- Experimental discovery programs

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11. Recommended Citation / Description

SSTAC is a geometry-driven sky survey planner designed for small-aperture observatories targeting asteroid and comet discovery. It prioritizes candidate fields based on solar elongation, altitude, twilight conditions, and observational constraints, providing an operational framework for efficient sky survey planning.

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12. Final Remarks

SSTAC bridges the gap between:

- Random sky survey
- Professional survey optimization systems

By embedding astronomical reasoning into a usable tool, it enables small observatories to perform **targeted, intelligent sky surveys** with meaningful discovery potential.